

BOWEN, AustraliaWMO: **943660**Lat: **20.0154S**Lon: **148.2140E**Elev: **8**StdP: **101.23**Time Zone: **10.00 (E10)**Period: **94-05**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	11.3	1.1	4.1	17.0	3.6	4.9	17.1	12.3	21.7	11.2	21.6	3.0	230	0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.5	32.6	25.8	31.9	25.4	31.3	25.1	27.7	30.8	27.2	30.2	26.7	29.7	6.3	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.9	22.6	29.7	26.3	21.8	29.2	25.8	21.1	28.8	88.6	31.0	86.0	30.1	83.8	30.0	31.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.6	10.4	9.8	5.6	34.6	1.0	1.5	4.9	35.6	4.4	36.5	3.8	37.3	3.1	38.4		
(5)			3.8	29.2	1.1	1.3	3.0	30.2	2.4	31.0	1.7	31.7	0.9	32.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.0	27.7	27.7	26.9	25.3	22.2	20.3	18.9	19.8	21.5	24.5	26.3	27.3
	DBStd	3.63	1.38	1.34	1.17	1.41	2.30	2.42	2.36	2.15	1.93	1.39	1.20	1.49
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	50	0	0	0	0	3	12	22	11	2	0	0	0
	CDD10.0	5114	549	496	523	458	377	308	277	304	346	451	488	537
	CDD18.3	2122	291	263	264	208	122	70	41	57	97	192	238	278
	CDH23.3	18492	3253	2898	2619	1540	617	184	104	207	590	1393	2151	2936
	CDH26.7	5392	1203	1034	824	354	59	1	3	5	53	277	590	988
Wind	WSAvg	5.3	5.5	5.4	6.0	5.9	5.2	5.4	5.0	5.1	4.6	5.1	5.3	5.2
Precipitation	PrecAvg	883	200	216	106	51	36	27	14	20	14	19	45	133
	PrecMax	1781	487	704	493	210	245	168	88	144	100	102	299	852
	PrecMin	391	37	3	7	0	1	1	0	0	0	0	0	14
	PrecStd	380	133	197	118	63	47	38	20	35	23	24	60	168
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.7	33.9	32.3	31.1	29.0	26.9	26.7	27.3	30.0	30.7	31.8	33.1
		MCWB	26.7	26.2	25.8	23.9	22.4	21.2	21.4	18.7	17.1	23.5	24.3	26.6
	2%	DB	32.6	32.5	31.4	30.2	28.0	26.1	25.3	26.1	27.7	29.5	30.7	31.8
		MCWB	26.2	25.7	25.0	23.8	21.9	20.7	19.9	19.3	20.9	23.3	24.6	25.6
	5%	DB	31.8	31.7	30.9	29.6	27.3	25.3	24.5	25.3	27.1	28.8	30.2	31.1
		MCWB	25.7	25.6	24.6	23.6	21.5	20.2	19.1	19.0	20.5	22.8	24.1	25.2
	10%	DB	31.1	31.0	30.2	28.9	26.5	24.6	23.7	24.6	26.4	28.2	29.6	30.5
		MCWB	25.2	25.4	24.3	23.2	21.0	19.6	18.4	18.7	20.1	22.2	23.7	24.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.6	28.0	27.6	26.0	24.9	24.1	24.0	23.2	24.0	26.0	28.0	28.6
		MCDB	32.2	31.2	30.8	29.5	27.3	25.3	24.4	25.2	27.3	29.4	30.6	31.5
	2%	WB	27.7	27.4	26.8	25.2	23.8	22.5	21.5	21.7	23.1	25.0	26.4	27.3
		MCDB	31.0	30.5	29.9	28.4	26.3	24.6	23.9	24.2	26.1	28.1	29.4	30.3
	5%	WB	27.1	26.9	26.1	24.8	22.8	21.6	20.4	20.7	22.2	24.2	25.5	26.6
		MCDB	30.1	30.0	29.1	27.9	25.8	24.1	23.2	23.6	25.6	27.1	28.8	29.8
	10%	WB	26.5	26.4	25.4	24.2	22.0	20.7	19.6	20.1	21.3	23.5	24.8	26.0
		MCDB	29.6	29.5	28.6	27.6	25.4	23.4	22.7	23.1	25.0	26.7	28.2	29.1
Mean Daily Temperature Range	5% DB	MDBR	6.5	6.1	6.8	7.1	8.5	8.3	9.9	9.7	10.1	7.8	6.8	6.3
		MCDBR	7.0	7.4	7.7	7.9	8.8	8.4	9.9	10.4	10.1	8.2	7.3	6.6
		MCWBR	2.7	2.7	3.0	3.4	4.2	4.3	5.2	5.2	4.8	3.5	3.3	2.6
	5% WB	MCDBR	6.1	6.3	6.5	7.3	8.3	6.7	8.1	8.9	8.6	7.4	6.7	6.1
MCWBR		3.0	2.8	3.3	3.7	5.2	4.0	5.0	5.6	4.6	3.8	4.4	2.9	
Clear-Sky Solar Irradiance	taub	0.405	0.399	0.386	0.364	0.347	0.334	0.321	0.330	0.359	0.385	0.390	0.401	
	taud	2.511	2.535	2.555	2.604	2.607	2.630	2.649	2.608	2.516	2.469	2.495	2.492	
	Ebn at Noon	940	935	922	906	885	879	904	927	934	937	949	944	
	Edh at Noon	115	110	104	92	85	80	81	90	106	116	116	117	
All-Sky Solar Radiation	RadAvg	6.41	6.09	5.61	5.03	4.33	3.89	4.37	5.21	6.28	6.93	7.04	7.03	
	RadStd	0.56	0.66	0.52	0.45	0.29	0.36	0.29	0.32	0.36	0.39	0.79	0.67	